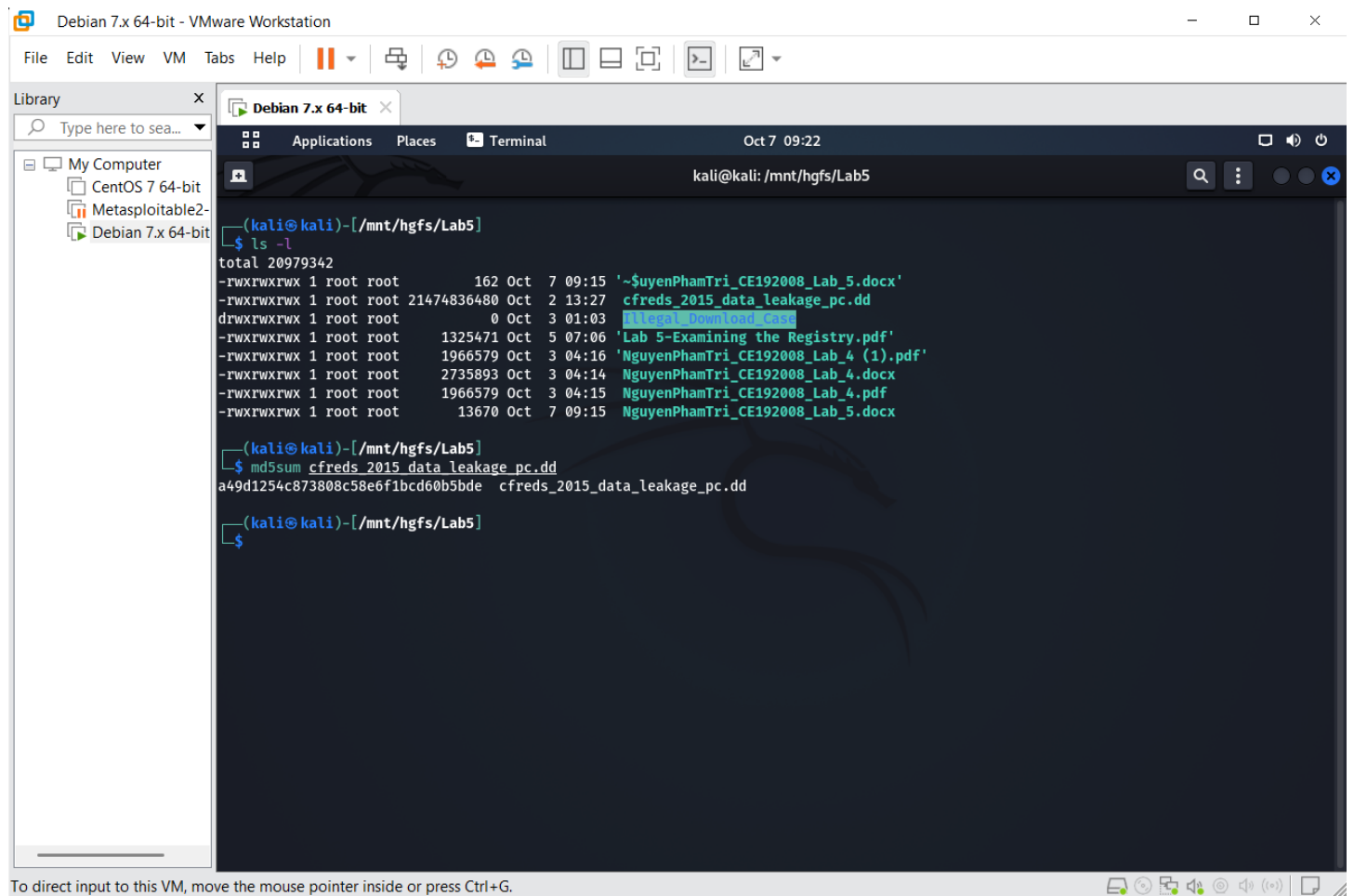


## LAB 5



Debian 7.x 64-bit - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- CentOS 7 64-bit
- Metasploitable2
- Debian 7.x 64-bit

Debian 7.x 64-bit

Applications Places Terminal

Oct 7 09:22

kali@kali: /mnt/hgfs/Lab5

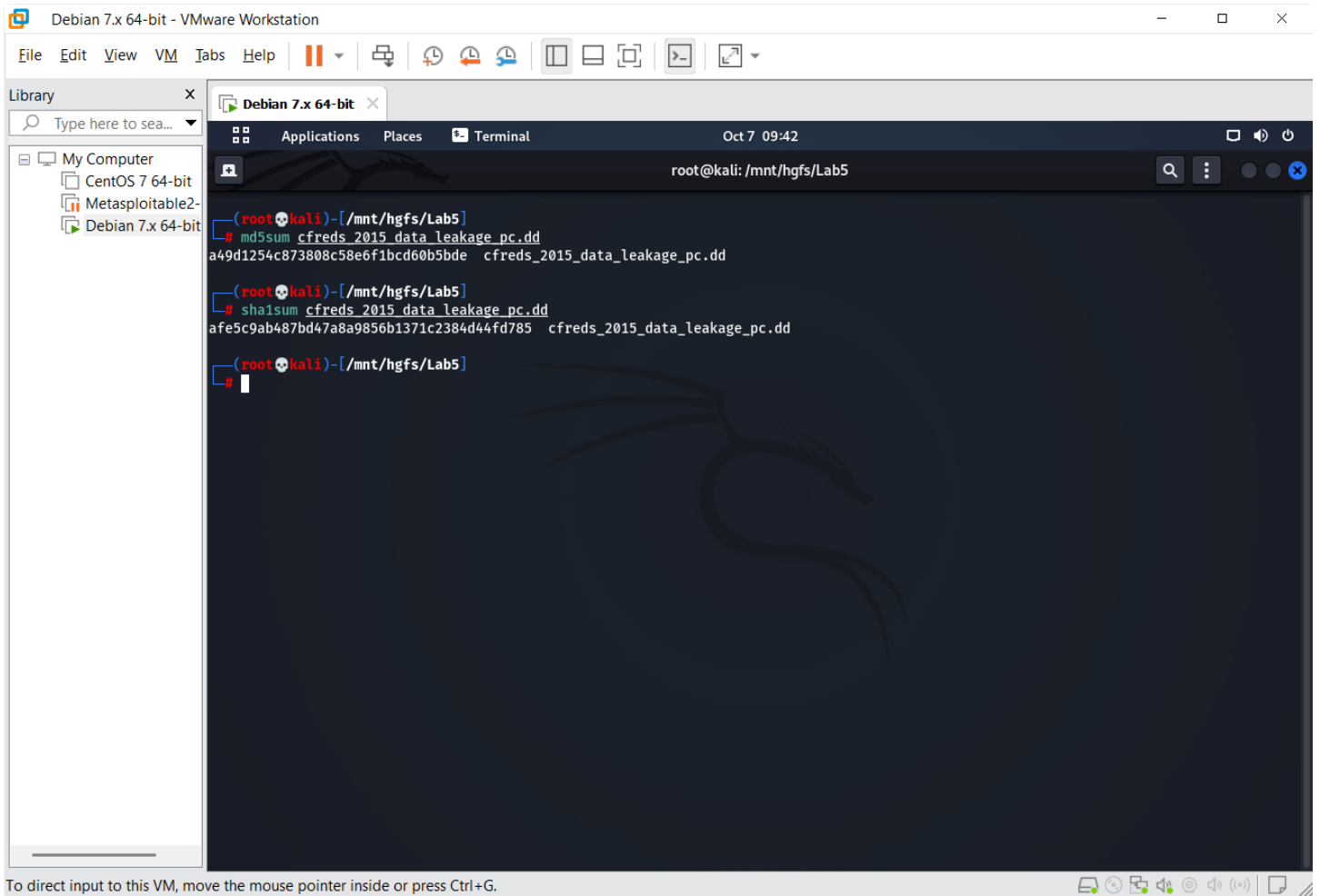
```
(kali@kali)-[/mnt/hgfs/Lab5]
$ ls -l
total 20979342
-rwxrwxrwx 1 root root 162 Oct 7 09:15 '~$uyenPhamTri_CE192008_Lab_5.docx'
-rwxrwxrwx 1 root root 21474836480 Oct 2 13:27 cfreds_2015_data_leakage_pc.dd
drwxrwxrwx 1 root root 0 Oct 3 01:03 illegal_download_cases
-rwxrwxrwx 1 root root 1325471 Oct 5 07:06 'Lab 5-Examining the Registry.pdf'
-rwxrwxrwx 1 root root 1966579 Oct 3 04:16 'NguyenPhamTri_CE192008_Lab_4 (1).pdf'
-rwxrwxrwx 1 root root 2735893 Oct 3 04:14 NguyenPhamTri_CE192008_Lab_4.docx
-rwxrwxrwx 1 root root 1966579 Oct 3 04:15 NguyenPhamTri_CE192008_Lab_4.pdf
-rwxrwxrwx 1 root root 13670 Oct 7 09:15 NguyenPhamTri_CE192008_Lab_5.docx

(kali@kali)-[/mnt/hgfs/Lab5]
$ md5sum cfreds_2015_data_leakage_pc.dd
a49d1254c873808c58e6f1bcd60b5bde cfreds_2015_data_leakage_pc.dd

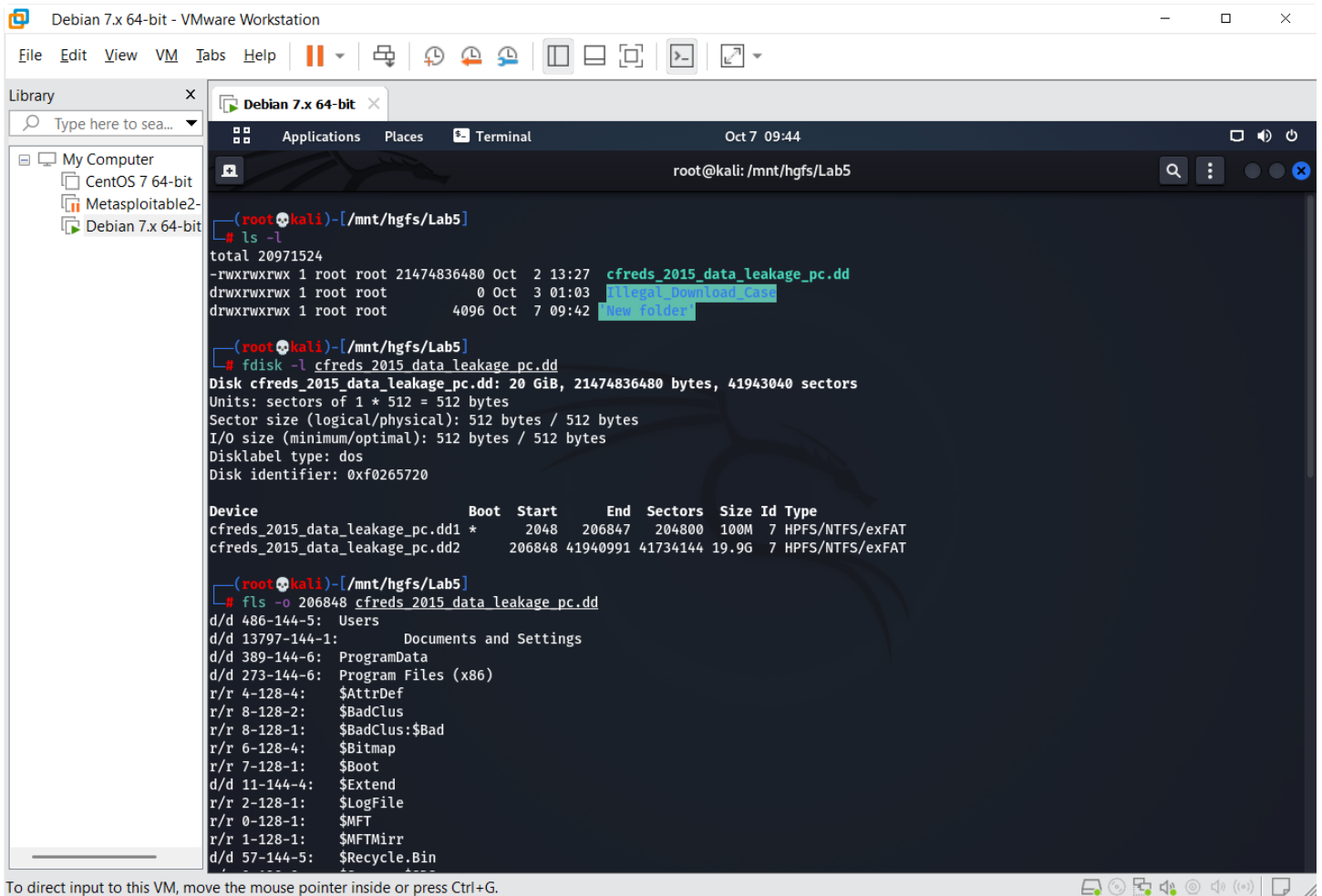
(kali@kali)-[/mnt/hgfs/Lab5]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

ảnh trên: tải file ảnh sử dụng cho bài lab



ảnh trên: tạo hash md5 và sha1 cho file ảnh



```
Debian 7.x 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
CentOS 7 64-bit
Metasploitable2-
Debian 7.x 64-bit

Debian 7.x 64-bit
Applications Places Terminal
Oct 7 09:44
root@kali: /mnt/hgfs/Lab5

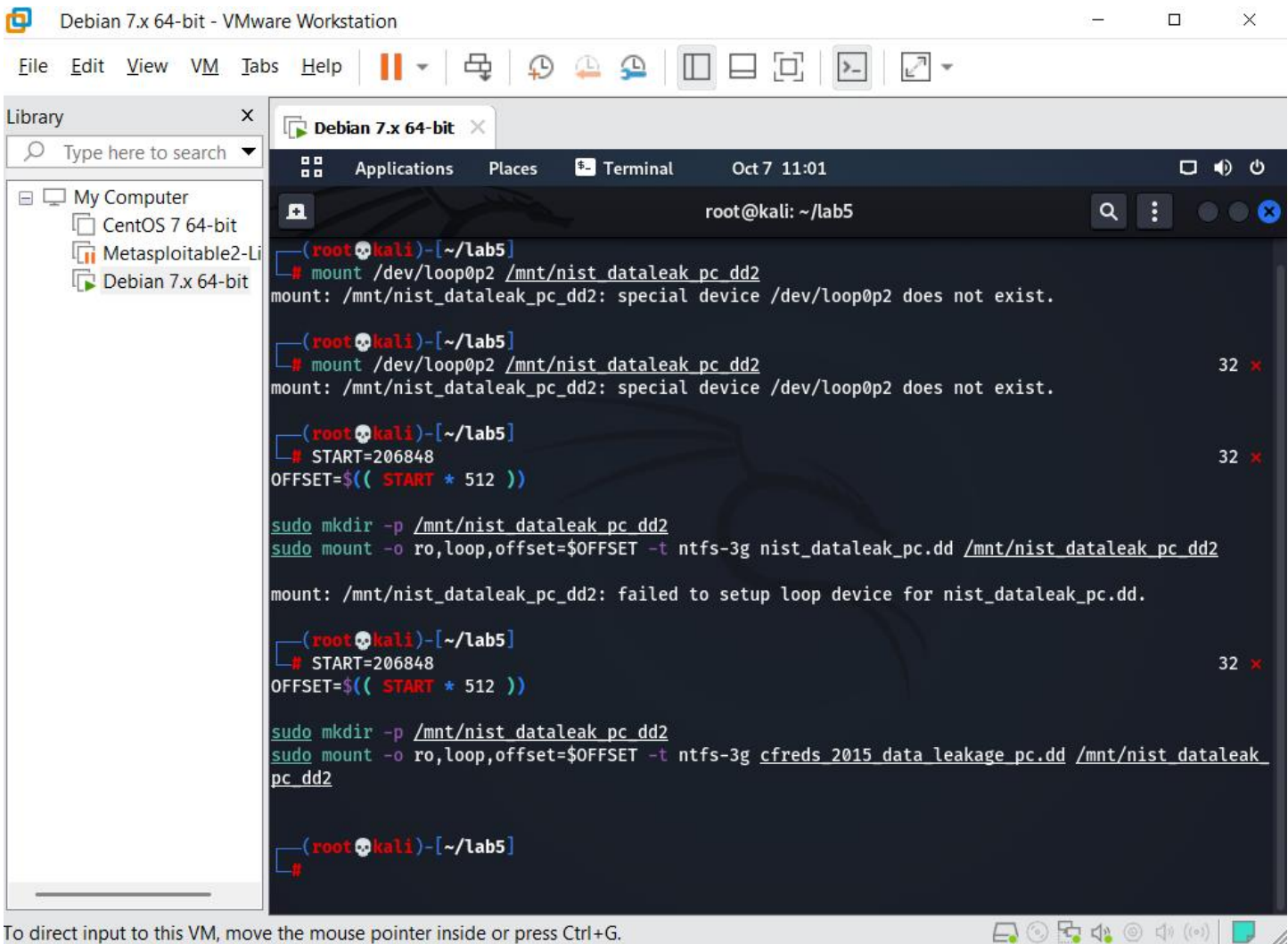
(root@kali)-[/mnt/hgfs/Lab5]
# ls -l
total 20971524
-rwxrwxrwx 1 root root 21474836480 Oct 2 13:27 cfreds_2015_data_leakage_pc.dd
drwxrwxrwx 1 root root 0 Oct 3 01:03 Illegal_Download_Case
drwxrwxrwx 1 root root 4096 Oct 7 09:42 New folder

(root@kali)-[/mnt/hgfs/Lab5]
# fdisk -l cfreds_2015_data_leakage_pc.dd
Disk cfreds_2015_data_leakage_pc.dd: 20 GiB, 21474836480 bytes, 41943040 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf0265720

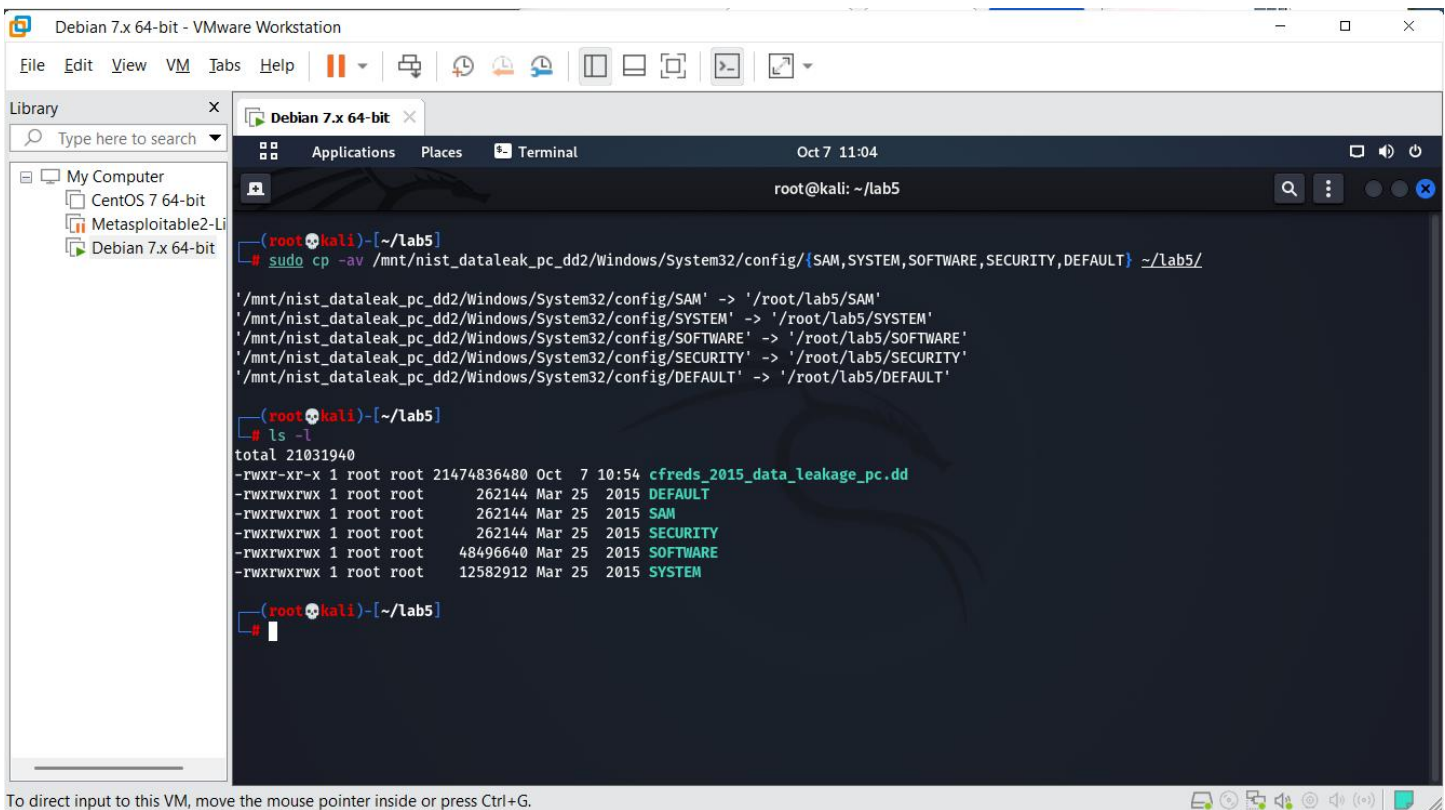
Device Boot Start End Sectors Size Id Type
cfreds_2015_data_leakage_pc.dd1 * 2048 206847 204800 100M 7 HPFS/NTFS/exFAT
cfreds_2015_data_leakage_pc.dd2 206848 41940991 41734144 19.9G 7 HPFS/NTFS/exFAT

(root@kali)-[/mnt/hgfs/Lab5]
# fls -o 206848 cfreds_2015_data_leakage_pc.dd
d/d 486-144-5: Users
d/d 13797-144-1: Documents and Settings
d/d 389-144-6: ProgramData
d/d 273-144-6: Program Files (x86)
r/r 4-128-4: $AttrDef
r/r 8-128-2: $BadClus
r/r 8-128-1: $BadClus:$Bad
r/r 6-128-4: $Bitmap
r/r 7-128-1: $Boot
d/d 11-144-4: $Extend
r/r 2-128-1: $LogFile
r/r 0-128-1: $MFT
r/r 1-128-1: $MFTMirr
d/d 57-144-5: $Recycle.Bin
```

ảnh trên: Kiểm tra phân vùng của hình ảnh DD bằng fdisk và liệt kê tên tệp/thư mục của ổ đĩa hệ thống

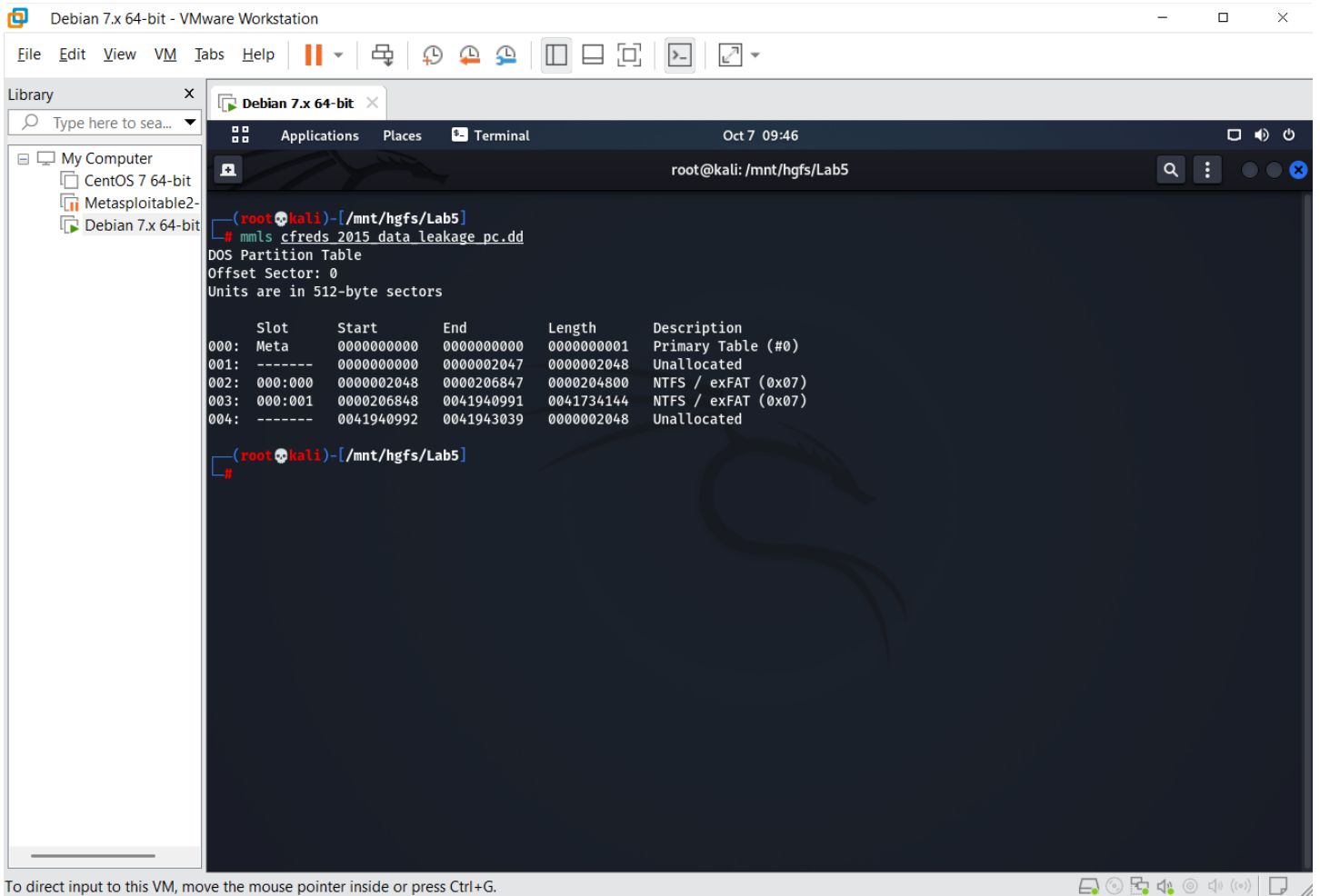


To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2 ảnh trên: mount ảnh đĩa và copy các file hệ thống để phân tích



```
(root@kali)-[/mnt/hgfs/Lab5]
# mmls cfreds_2015_data_leakage_pc.dd
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

    Slot    Start      End      Length  Description
000: Meta   0000000000  0000000000  0000000001 Primary Table (#0)
001: ----- 0000000000  0000002047  0000002048 Unallocated
002: 000:000 0000002048  0000206847  0000204800 NTFS / exFAT (0x07)
003: 000:001 0000206848  0041940991  0041734144 NTFS / exFAT (0x07)
004: ----- 0041940992  0041943039  0000002048 Unallocated

(root@kali)-[/mnt/hgfs/Lab5]
#
```

ảnh trên: Hiển thị các phân vùng và không gian chưa phân bổ bằng mmls

Debian 7.x 64-bit - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- CentOS 7 64-bit
- Metasploitable2
- Debian 7.x 64-bit

Debian 7.x 64-bit

Applications Places Terminal

Oct 7 09:47

root@kali: /mnt/hgfs/Lab5

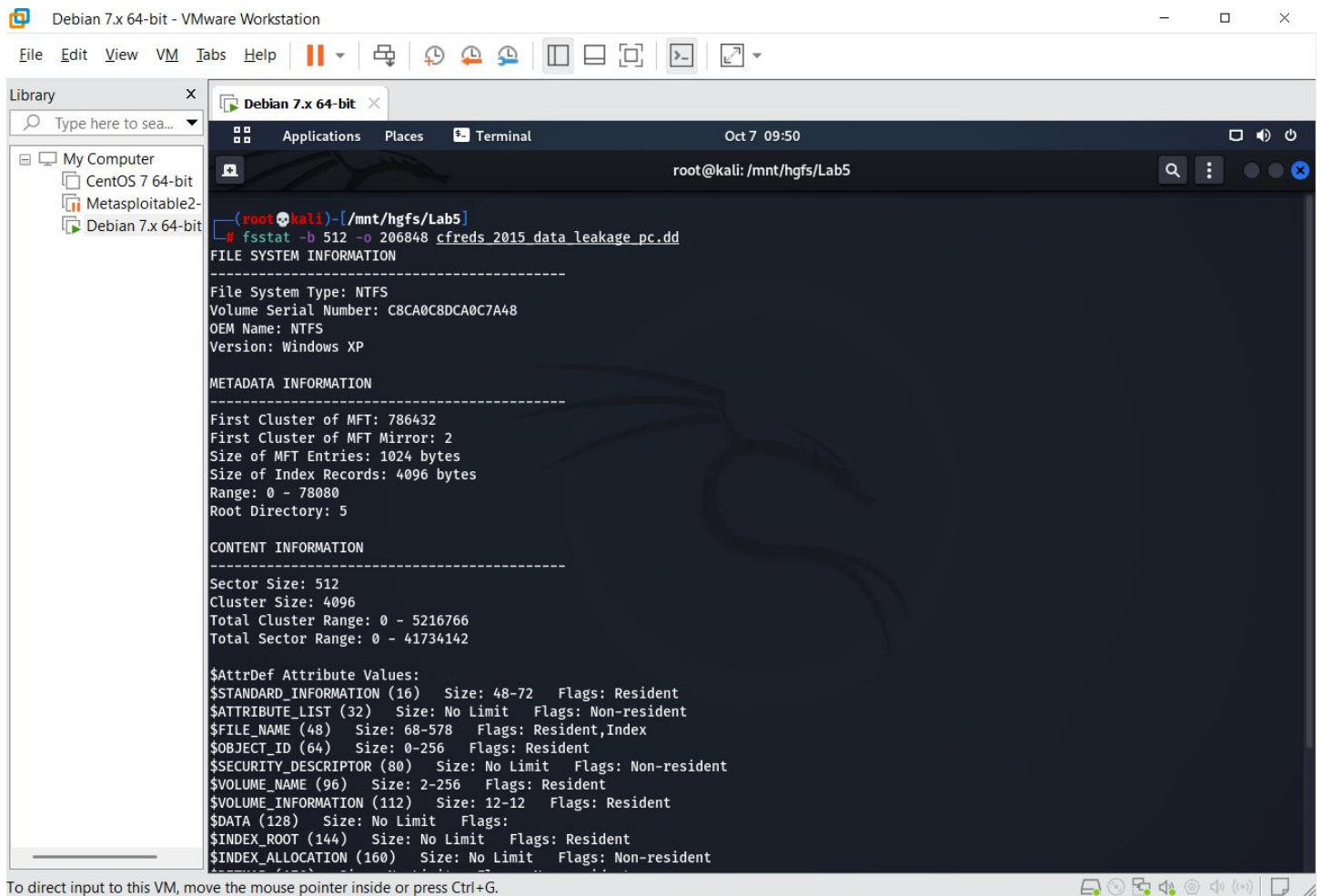
```
(root@kali)~[/mnt/hgfs/Lab5]
# fsstat -b 512 -o 2048 cfreds_2015_data_leakage_pc.dd
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: 4A180A15180A0125
OEM Name: NTFS
Volume Name: System Reserved
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 8533
First Cluster of MFT Mirror: 2
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 256
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 25598
Total Sector Range: 0 - 204798

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



Debian 7.x 64-bit - VMware Workstation

File Edit View VM Tabs Help

Library

Debian 7.x 64-bit

My Computer

- CentOS 7 64-bit
- Metasploitable2
- Debian 7.x 64-bit

Applications Places Terminal

Oct 7 09:50

root@kali: /mnt/hgfs/Lab5

```
(root@kali)~[/mnt/hgfs/Lab5]
# fsstat -b 512 -o 206848 cfreds_2015_data_leakage_pc.dd
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: C8CA0C8DCA0C7A48
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 786432
First Cluster of MFT Mirror: 2
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 78080
Root Directory: 5

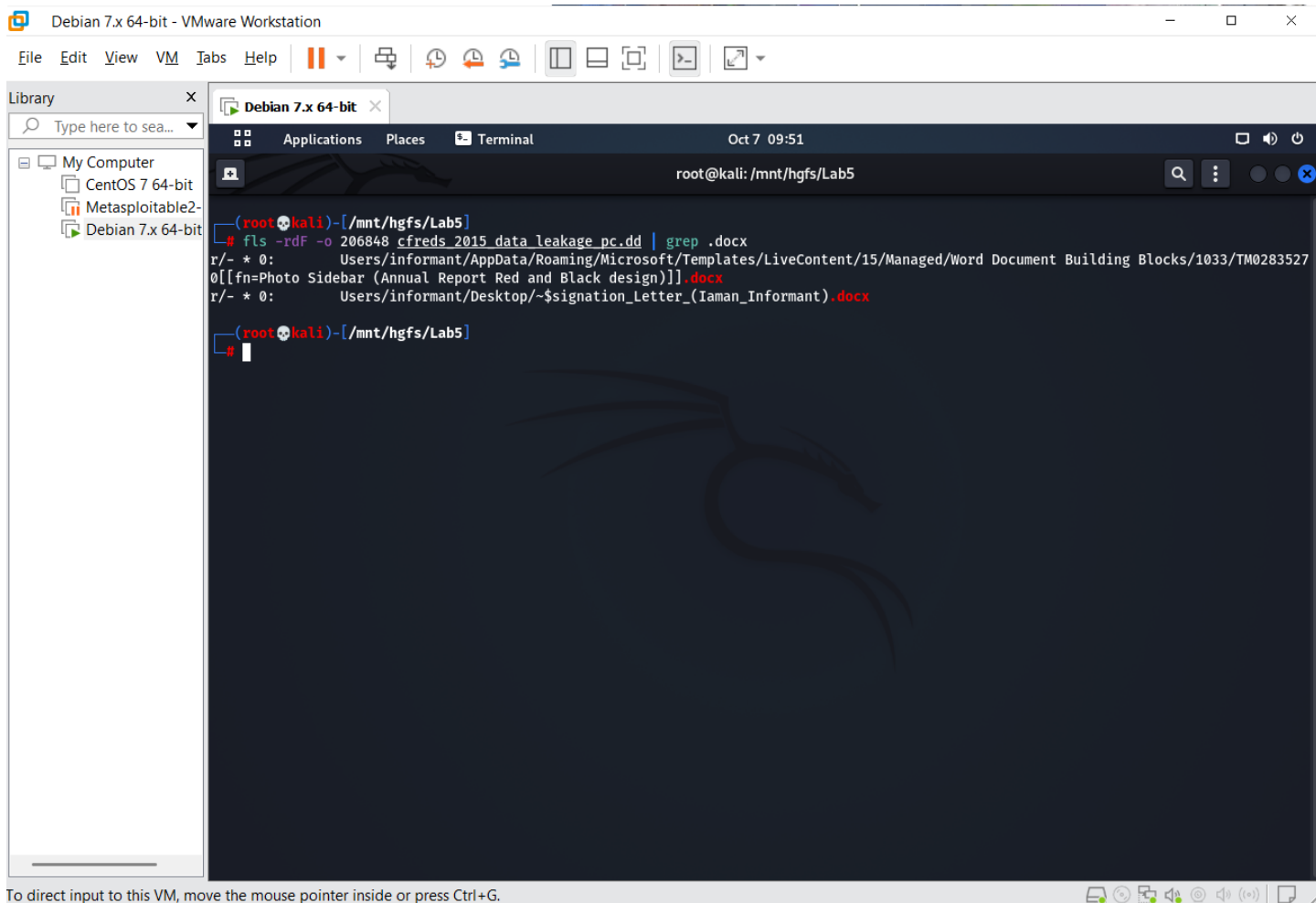
CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 5216766
Total Sector Range: 0 - 41734142

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2 ảnh trên: Hiển thị số liệu thống kê hệ thống tệp và thông tin siêu dữ liệu từ ảnh đĩa của 2 phân vùng

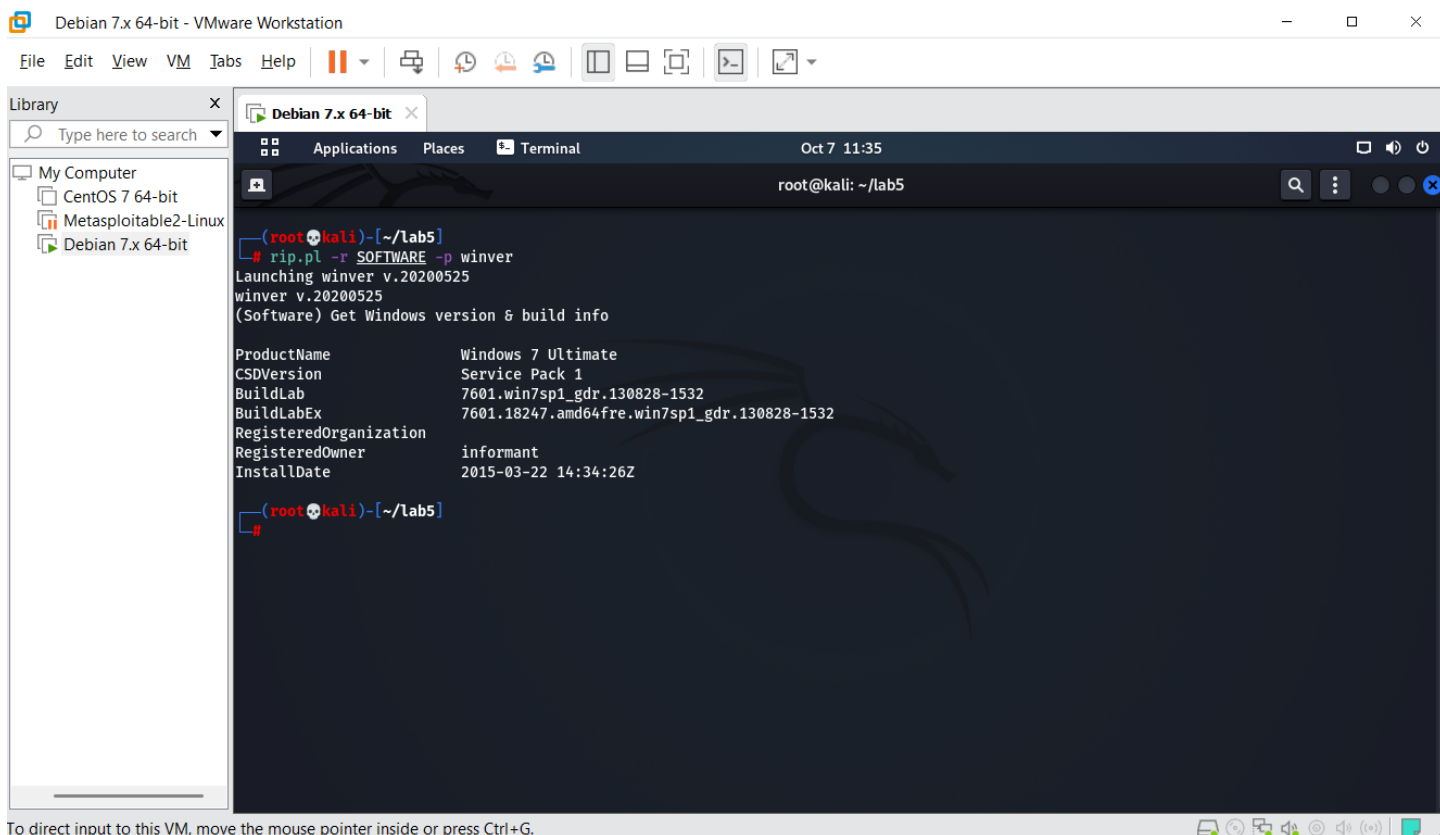




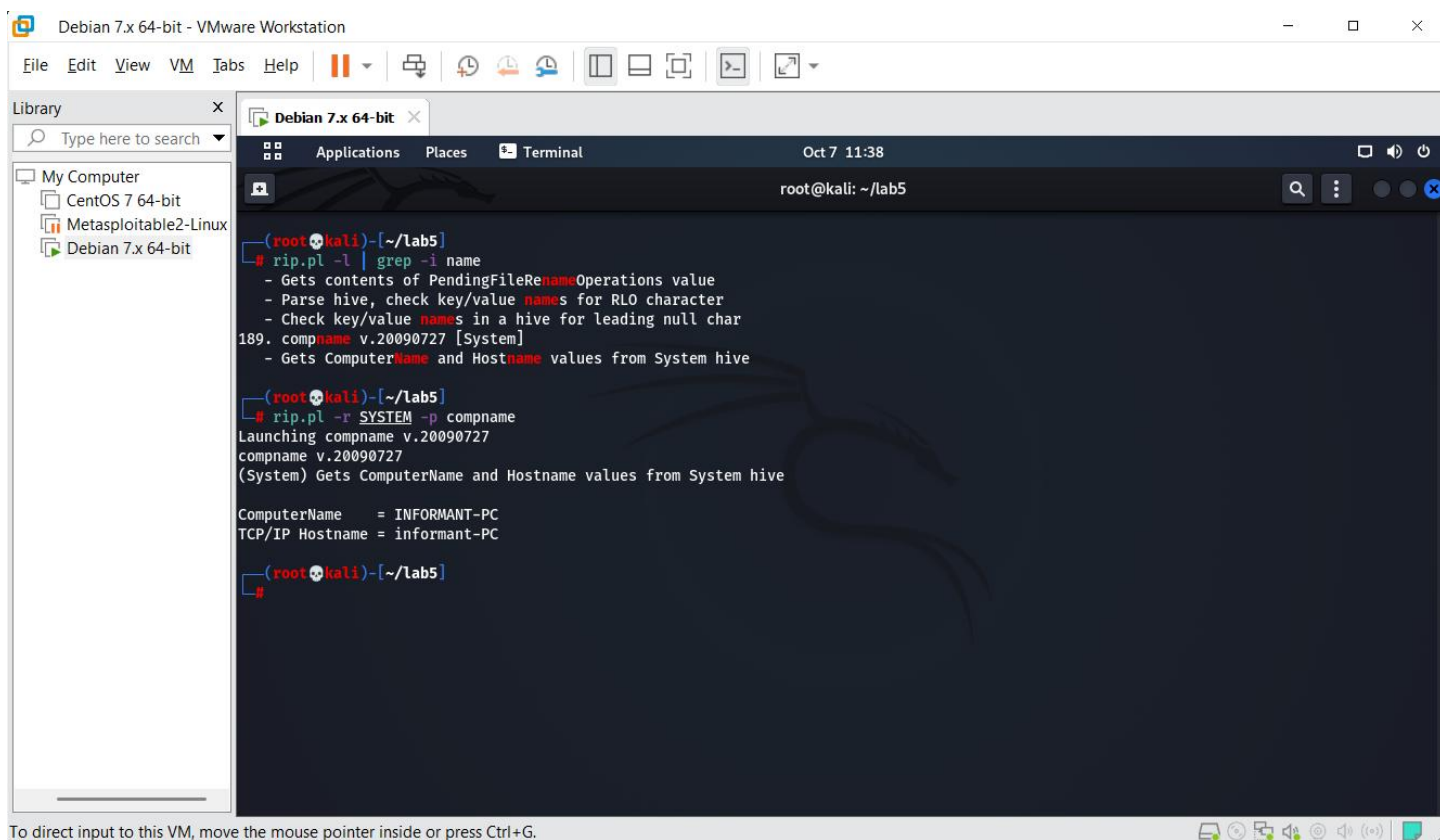
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Liệt kê tất cả các tệp .docx đã xóa trong toàn bộ phân vùng

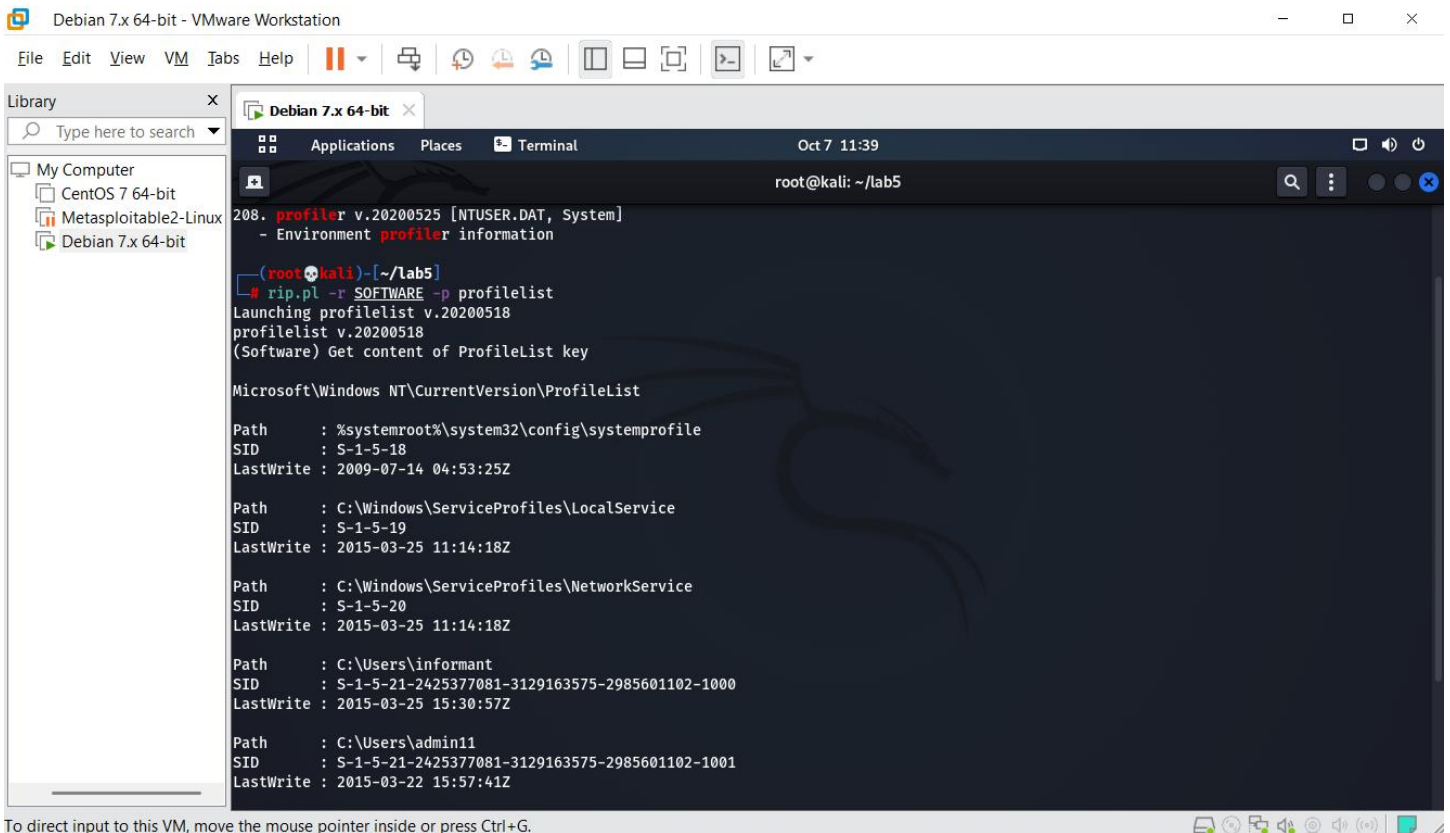




ảnh trên: Thông tin chi tiết về hệ điều hành đã cài đặt là gì? **Windows 7 Ultimate**



ảnh trên: What is the computer name? **INFORMANT-PC**



Debian 7.x 64-bit - VMware Workstation

File Edit View VM Tabs Help

Library

Debian 7.x 64-bit

Applications Places Terminal

Oct 7 11:39

root@kali: ~/lab5

```
208. profiler v.20200525 [NTUSER.DAT, System]
- Environment profiler information

(root@kali)~/lab5
# rip.pl -r SOFTWARE -p profilelist
Launching profilelist v.20200518
profilelist v.20200518
(Software) Get content of ProfileList key

Microsoft\Windows NT\CurrentVersion\ProfileList

Path      : %systemroot%\system32\config\systemprofile
SID       : S-1-5-18
LastWrite : 2009-07-14 04:53:25Z

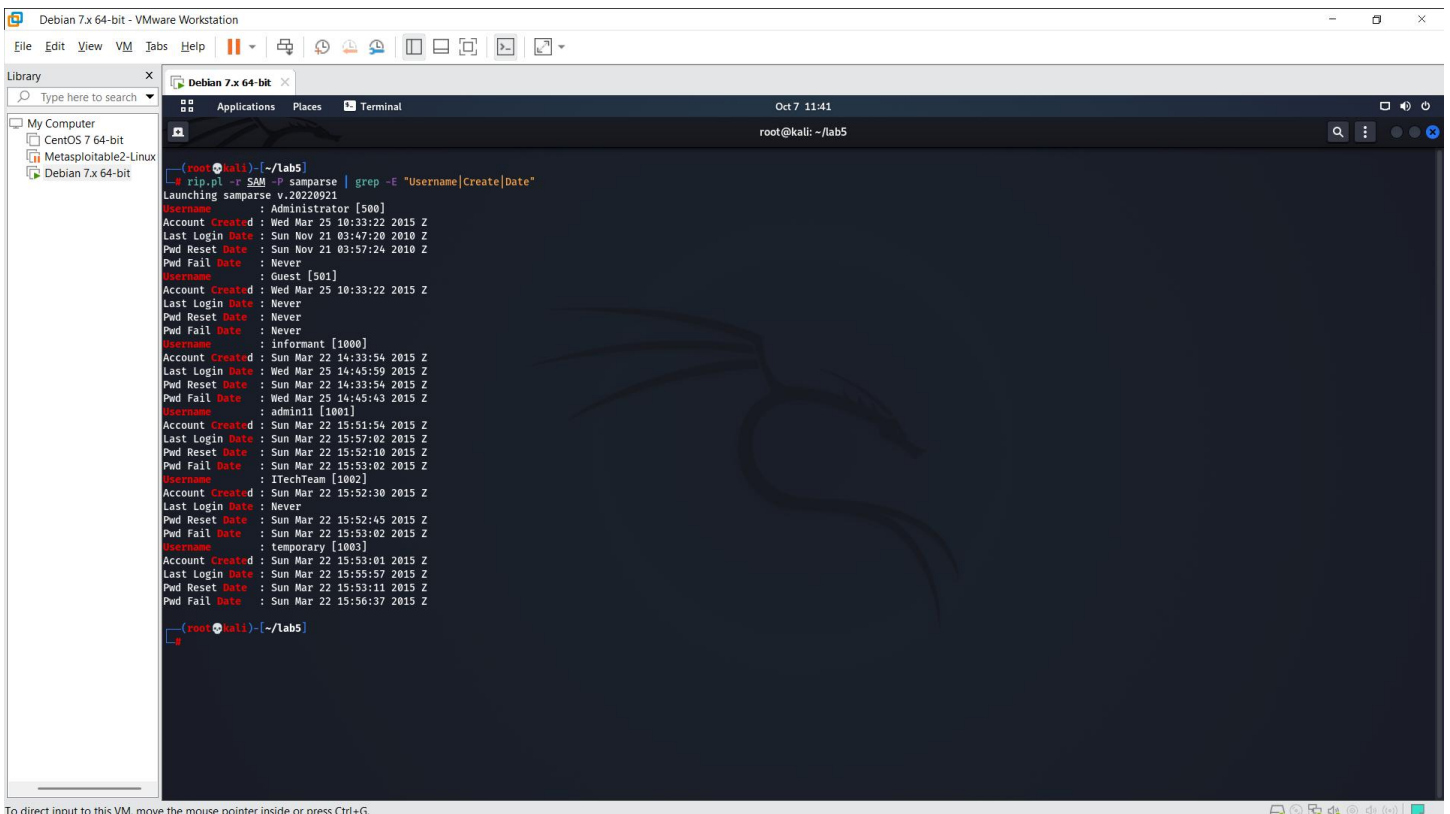
Path      : C:\Windows\ServiceProfiles\LocalService
SID       : S-1-5-19
LastWrite : 2015-03-25 11:14:18Z

Path      : C:\Windows\ServiceProfiles\NetworkService
SID       : S-1-5-20
LastWrite : 2015-03-25 11:14:18Z

Path      : C:\Users\informant
SID       : S-1-5-21-2425377081-3129163575-2985601102-1000
LastWrite : 2015-03-25 15:30:57Z

Path      : C:\Users\admin11
SID       : S-1-5-21-2425377081-3129163575-2985601102-1001
LastWrite : 2015-03-22 15:57:41Z
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



Debian 7.x 64-bit - VMware Workstation

File Edit View VM Tabs Help

Library

Debian 7.x 64-bit

Applications Places Terminal

Oct 7 11:41

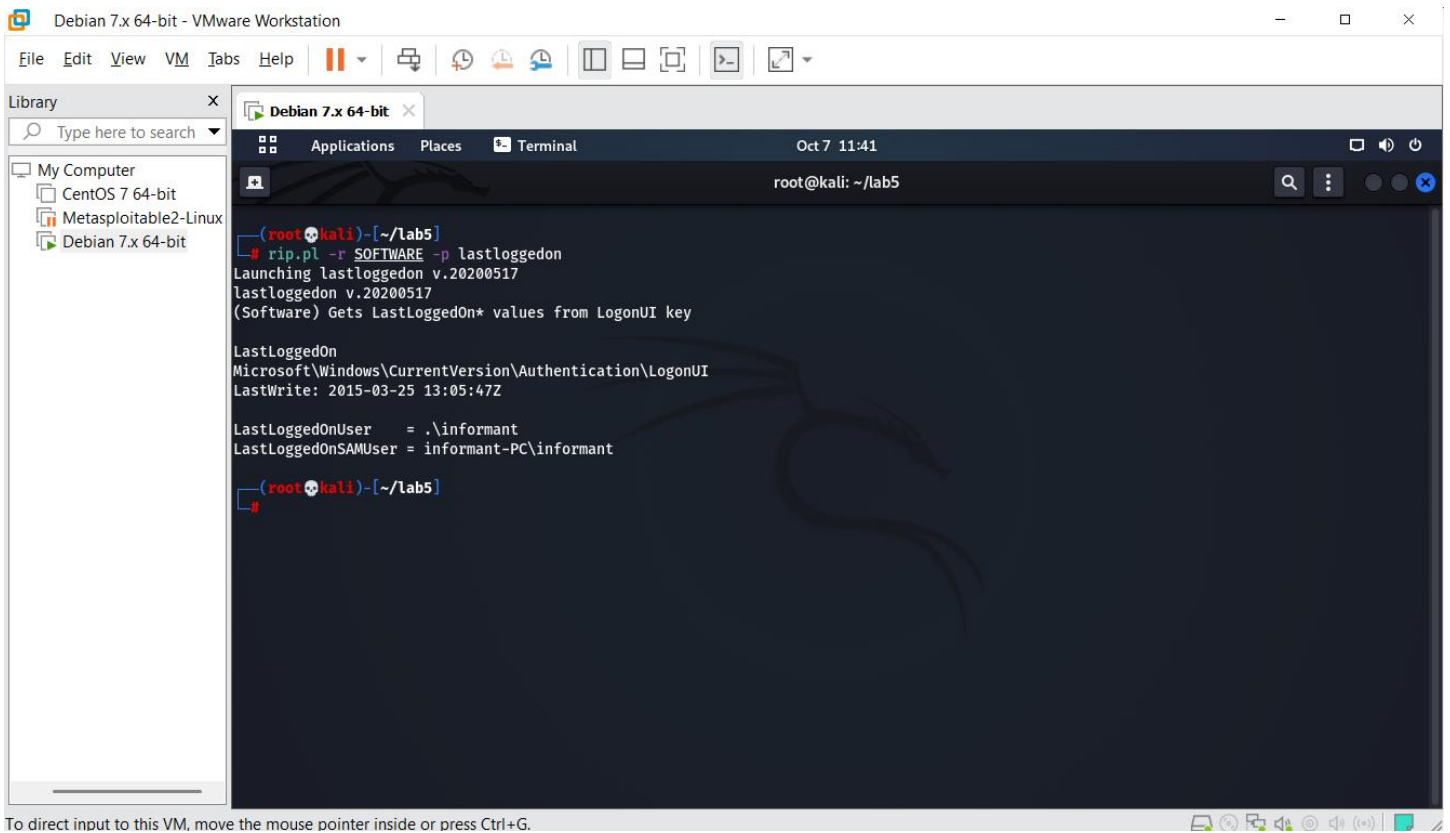
root@kali: ~/lab5

```
(root@kali)~/lab5
# rip.pl -r SAM -P samparse | grep -E "Username|create|date"
Launching samparse v.20220921
Username      : Administrator [500]
Account Created : Wed Mar 25 10:33:22 2015 Z
Last Login Date : Sun Nov 21 03:47:20 2010 Z
Pwd Reset Date  : Sun Nov 21 03:57:24 2010 Z
Pwd Fail Date   : Never
Username       : Guest [501]
Account Created : Wed Mar 25 10:33:22 2015 Z
Last Login Date : Never
Pwd Reset Date  : Never
Pwd Fail Date   : Never
Username       : informant [1000]
Account Created : Sun Mar 22 14:33:54 2015 Z
Last Login Date : Wed Mar 25 14:45:59 2015 Z
Pwd Reset Date  : Sun Mar 22 14:33:54 2015 Z
Pwd Fail Date   : Wed Mar 25 14:45:43 2015 Z
Username       : admin11 [1001]
Account Created : Sun Mar 22 15:51:54 2015 Z
Last Login Date : Sun Mar 22 15:57:02 2015 Z
Pwd Reset Date  : Sun Mar 22 15:52:10 2015 Z
Pwd Fail Date   : Sun Mar 22 15:53:02 2015 Z
Username       : ITechTeam [1002]
Account Created : Sun Mar 22 15:52:30 2015 Z
Last Login Date : Never
Pwd Reset Date  : Sun Mar 22 15:52:45 2015 Z
Pwd Fail Date   : Sun Mar 22 15:53:02 2015 Z
Username       : Temporary [1003]
Account Created : Sun Mar 22 15:53:01 2015 Z
Last Login Date : Sun Mar 22 15:55:57 2015 Z
Pwd Reset Date  : Sun Mar 22 15:53:11 2015 Z
Pwd Fail Date   : Sun Mar 22 15:56:37 2015 Z

(root@kali)~/lab5
```

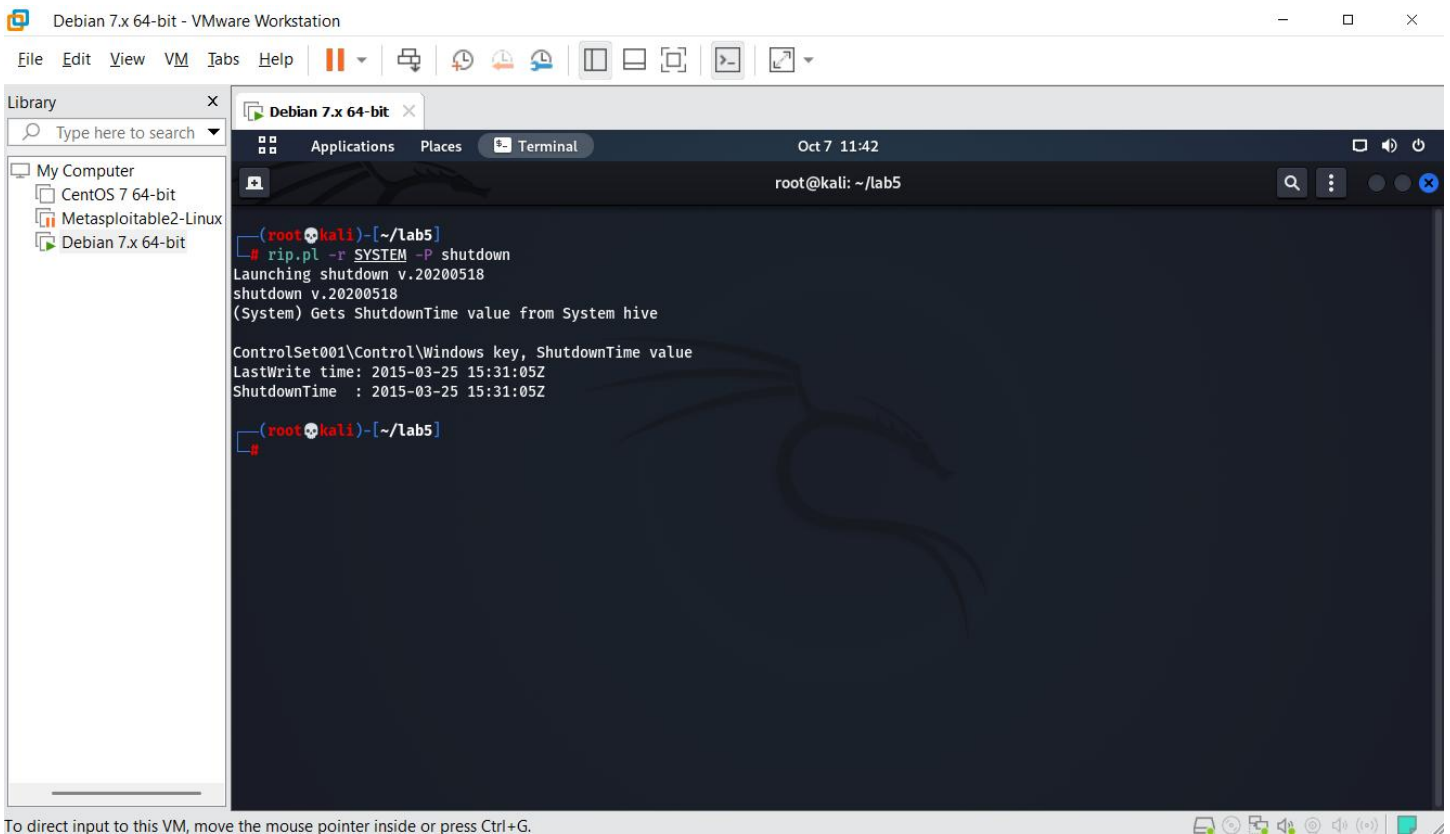
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2 ảnh trên: How many accounts does the system have? Admintrator, Guest, informant, admin11, ITechTeam, temporary



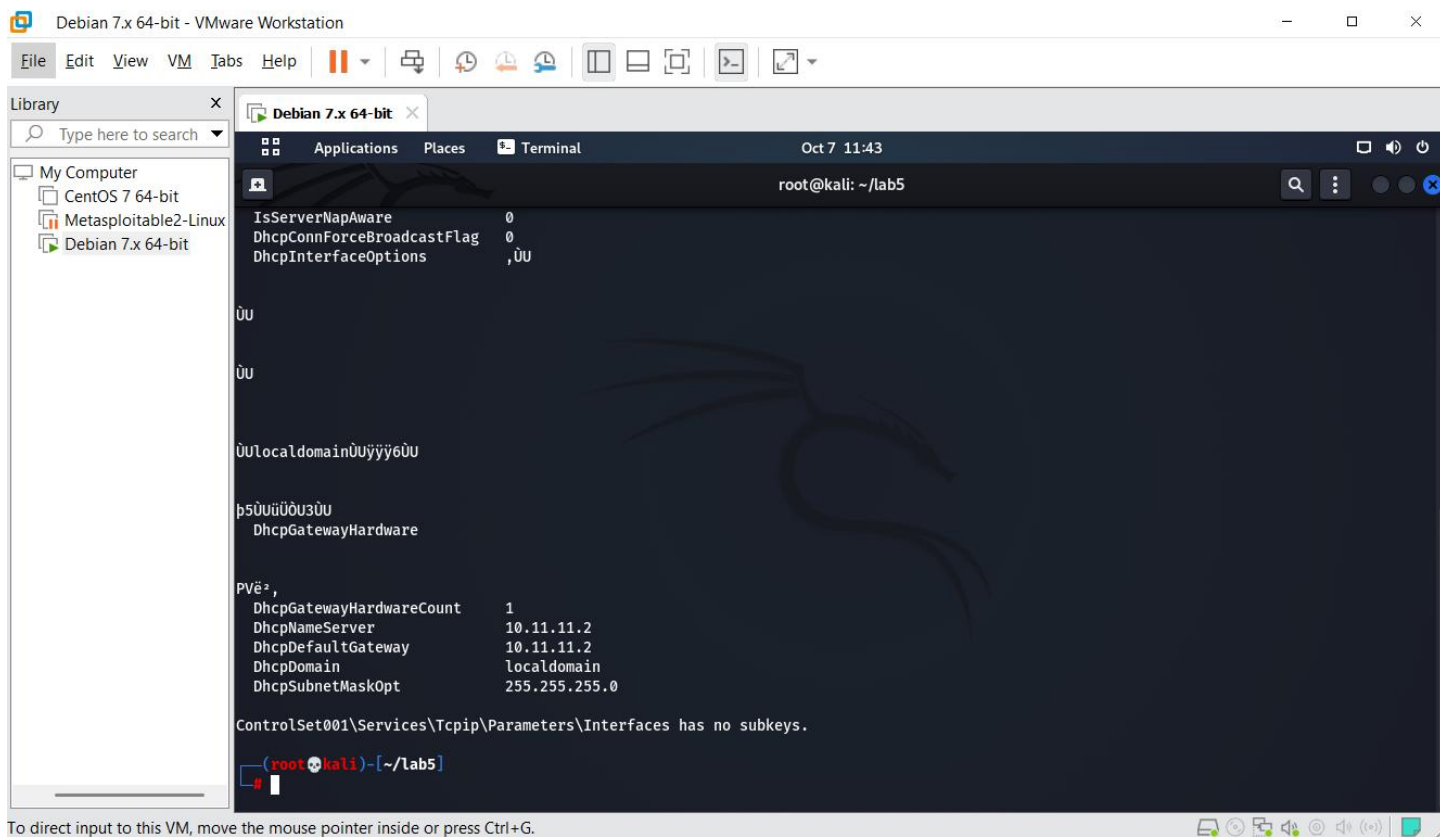
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

ảnh trên: Last logged-on user = informant



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

ảnh trên: When was the last recorded shutdown date/time? 2015-03-25 15:31:05Z



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

ảnh trên: Explain the information of network interface(s) with an IP address assigned by DHCP.

**DhcpNameServer**

**10.11.11.2**

**DhcpDefaultGateway**

**10.11.11.2**

**DhcpDomain localdomain**

**DhcpSubnetMaskOpt (Subnet Mask) 255.255.255.0**